

Ex. 4: OFDMA RESOURCES BLOCK Allocation

Code:

```
clc;
clear;
close all;
N_RB = 20;
N_users = 5;
base = floor(N_RB / N_users);
rem = mod(N_RB, N_users);
alloc_equal = zeros(1, N_RB);
idx = 1;
for u = 1:N_users
    take = base + (u <= rem);
    alloc_equal(idx:idx+take-1) = u;
    idx = idx + take;
end
alloc_rr = zeros(1, N_RB);
for rb = 1:N_RB
    alloc_rr(rb) = mod(rb-1, N_users) + 1;
end
figure;
subplot(2,1,1);
imagesc(alloc_equal);
colormap(parula(N_users));
cb1 = colorbar;
cb1.Ticks = 1:N_users;
cb1.TickLabels = {'U1','U2','U3','U4','U5'};
title('Equal Resource Block Allocation');
xlabel('Resource Block Index');
ylabel('Allocation');
subplot(2,1,2);
imagesc(alloc_rr);
colormap(parula(N_users));
cb2 = colorbar;
cb2.Ticks = 1:N_users;
cb2.TickLabels = {'U1','U2','U3','U4','U5'};
title('Round-Robin Resource Block Allocation');
xlabel('Resource Block Index');
ylabel('Allocation');
```

```

disp('Equal Allocation:');
disp(alloc_equal);
disp('Round-Robin Allocation:');
disp(alloc_rr);

```

